

Agentic AI for *Biomedical Data Science*

Hands-on portion

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LIVE DEMO

A Framework to Manage Context and Improve Results



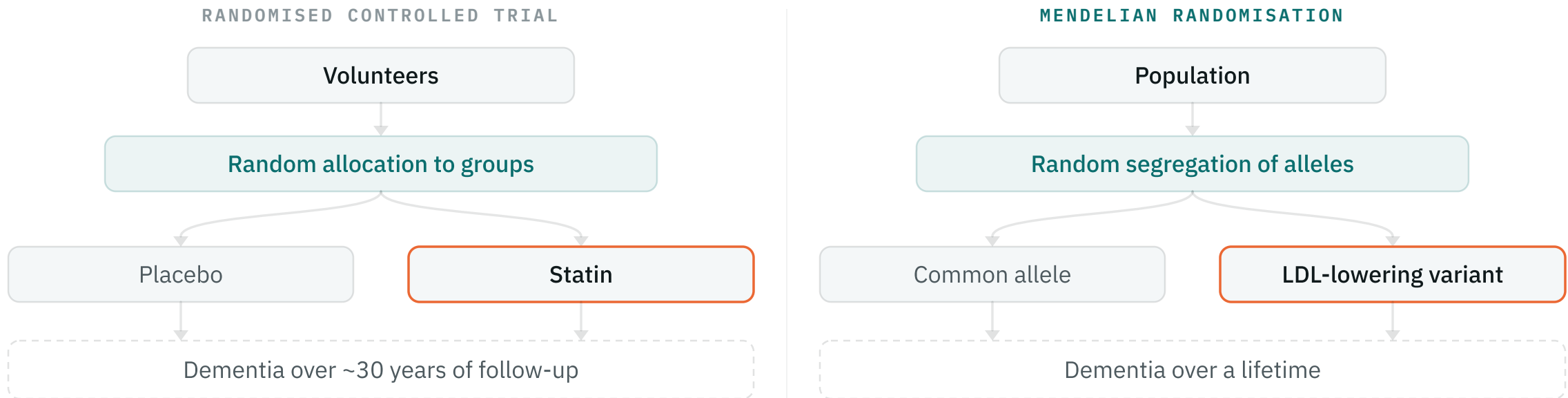
- **New chat sessions for each phase:** clears irrelevant information to improve model performance
- **Mirrors engineering and project-management workflows:** drives high-quality results

Do cholesterol-lowering drugs reduce the risk of *dementia*?

Reproducing a published paper with **Claude Code**

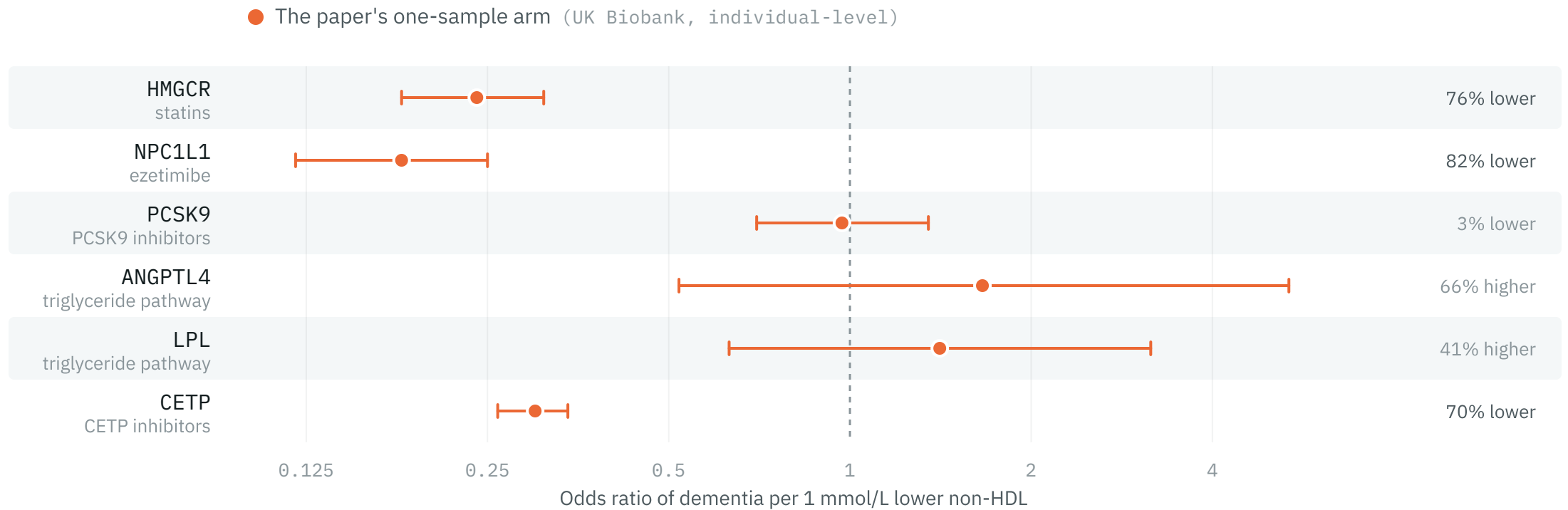
Nordestgaard LT, Hanson A, Sanderson E, et al. Cholesterol-lowering drug targets reduce risk of dementia: Mendelian randomization and meta-analyses of 1 million individuals. *Alzheimer's & Dementia* 2025;21:e70638. doi:10.1002/alz.70638

Mendelian randomisation gets causal evidence from observational data

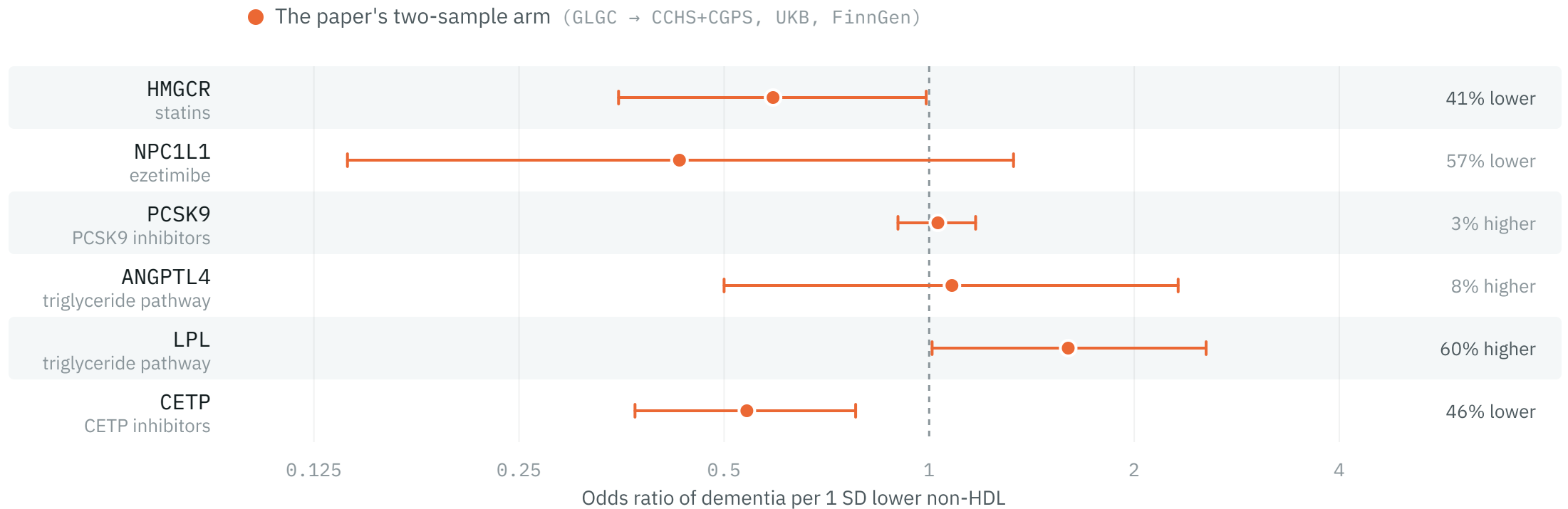


- If we know both effects — alleles on LDL, alleles on dementia — in the **same individuals: One-sample MR**
- If we know each effect from a **different study: Two-sample MR**

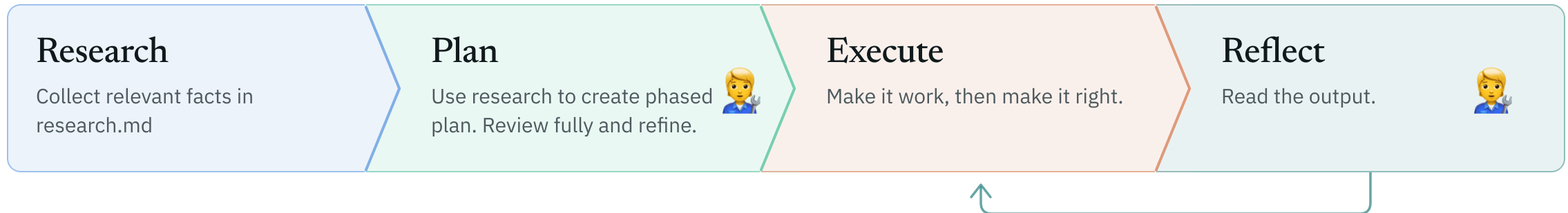
Variants in targets of cholesterol-lowering drugs reduce dementia risk



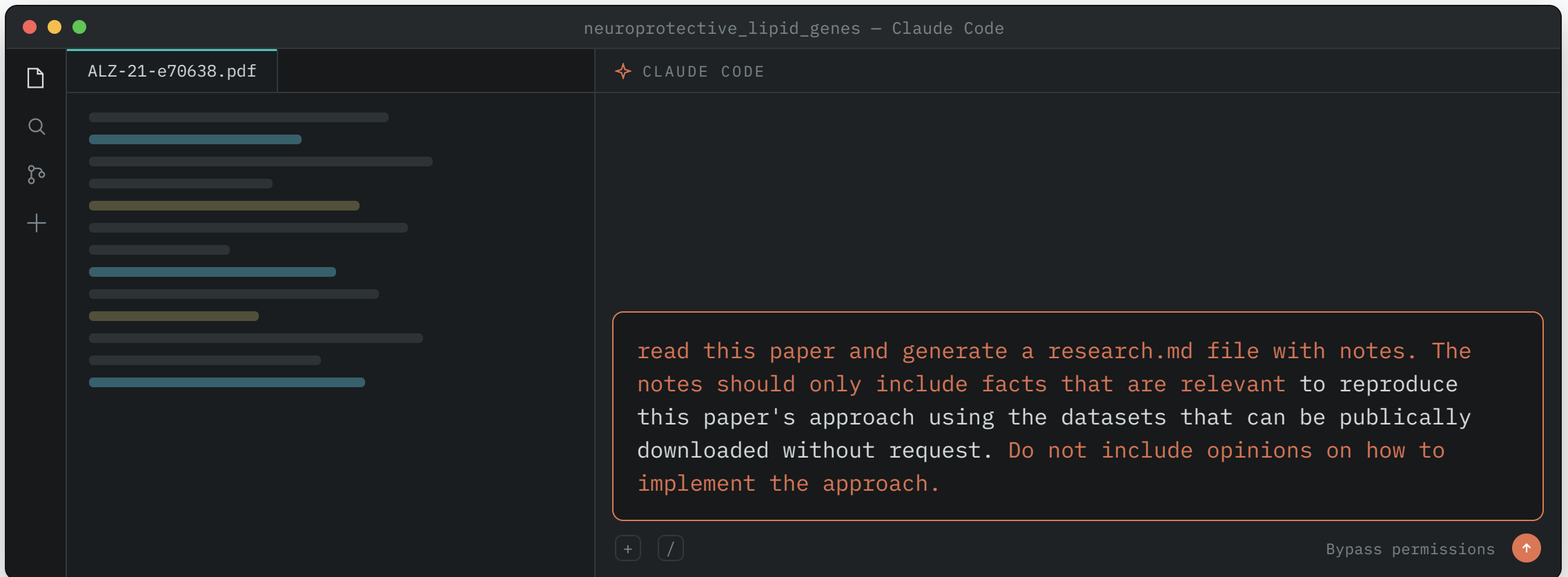
The two-sample arm is what we can reproduce



A Framework to Manage Context and Improve Results



Ask for facts, not opinions



```
# Reproducing the LDL-lowering to dementia analysis
:

### Datasets and releases
- GLGC 2021 – variant to LDL cholesterol
- FinnGen R12 – variant to dementia
...

### Choosing the variants
- a window around each drug-target gene
...

### How the estimate is calculated
...

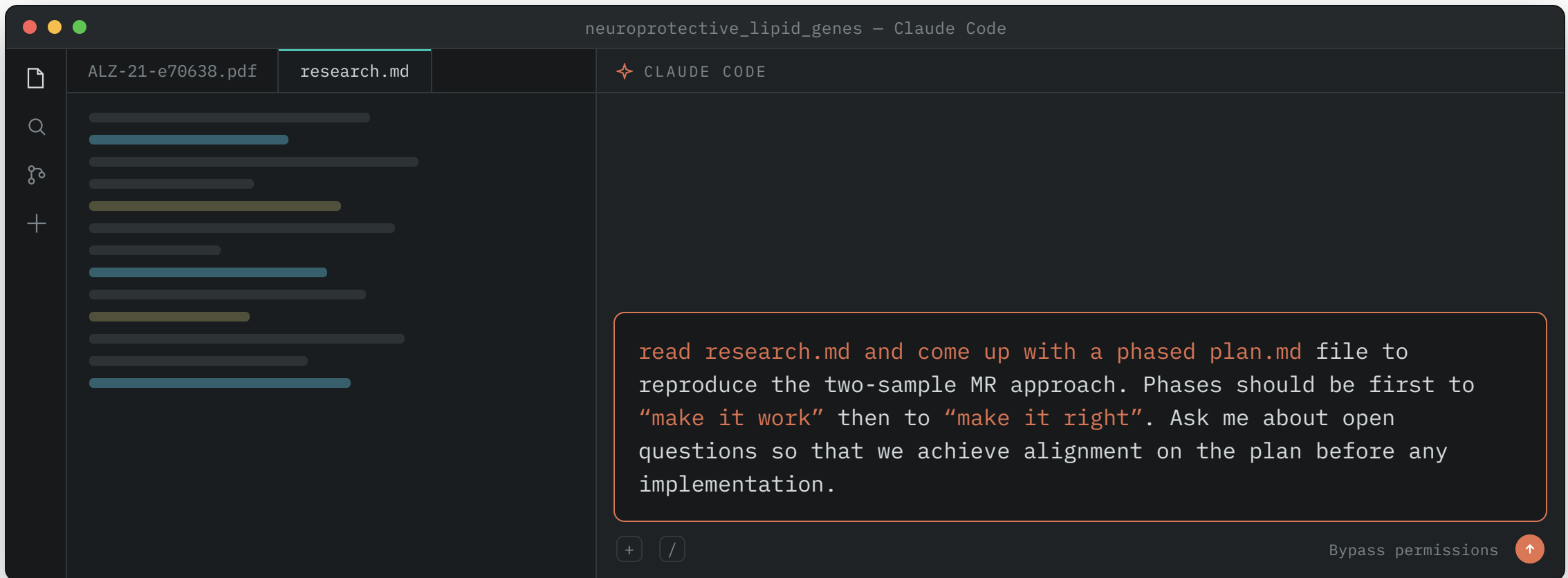
### The paper's own estimates
```

Two of five datasets are obtainable

DATASET	STATUS
UK Biobank	application required
Copenhagen ×2	cannot be shared
GLGC	open — variant → cholesterol
FinnGen	open — variant → disease

- **Details of approach** — six drug-target genes, non-HDL cholesterol, four dementia endpoints, heart disease as the positive control
- **Data location and release** — GLGC without UK Biobank, FinnGen
- **How to choose the variants** — the window around each gene, the paper's own rsIDs
- **How the estimate is calculated**
- **The paper's estimates to compare against** — for heart disease (positive control) and dementia

Ask for a phased plan based on research



We settled on four phases



neuroprotective_lipid_genes - Claude Code

plan.md Preview plan.md

Plan – reproducing the two-sample arm

...

Phases

PHASE	GOAL	DONE WHEN
0 · acquire	get the data	six gene regions cached, both sources
1 · make it work	works end to end	6 genes × 2 diseases, use rsIDs
2 · make it right	the paper's real method	our own variants, 5 diseases, assumption checks
3 · stretch	optional extension	more Alzheimer's cases

Review plan carefully before implementation

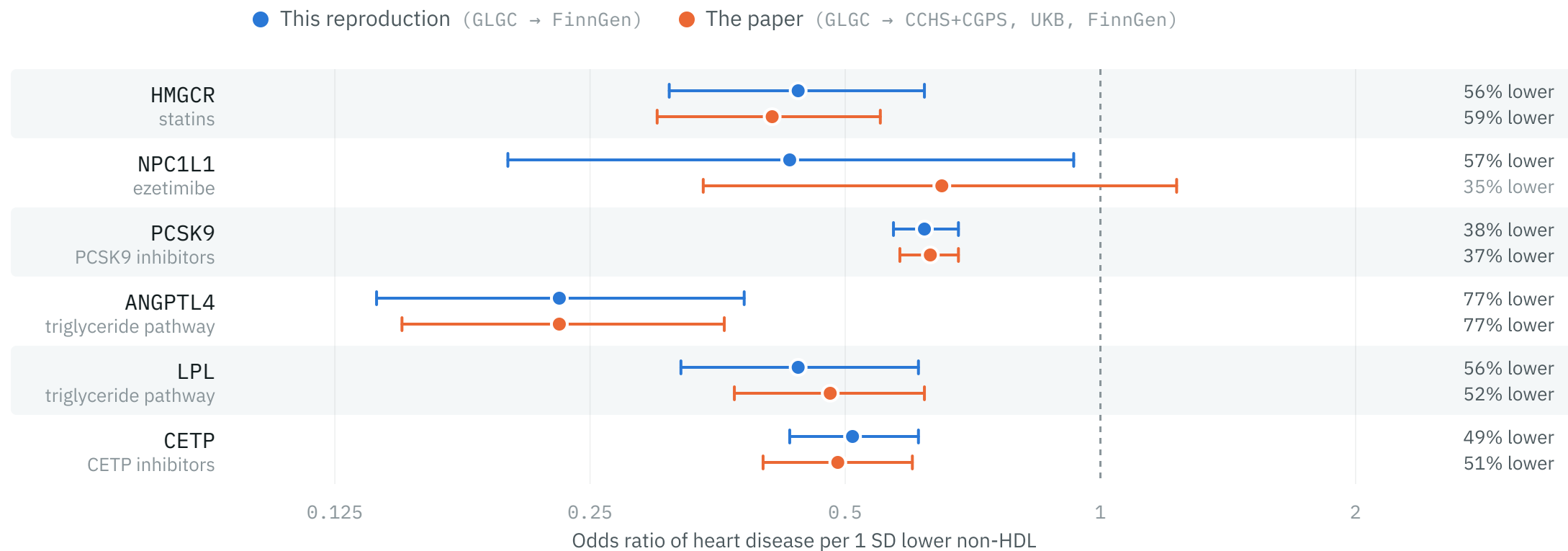
ALEX

Proceed with the phased implementation in plan.md

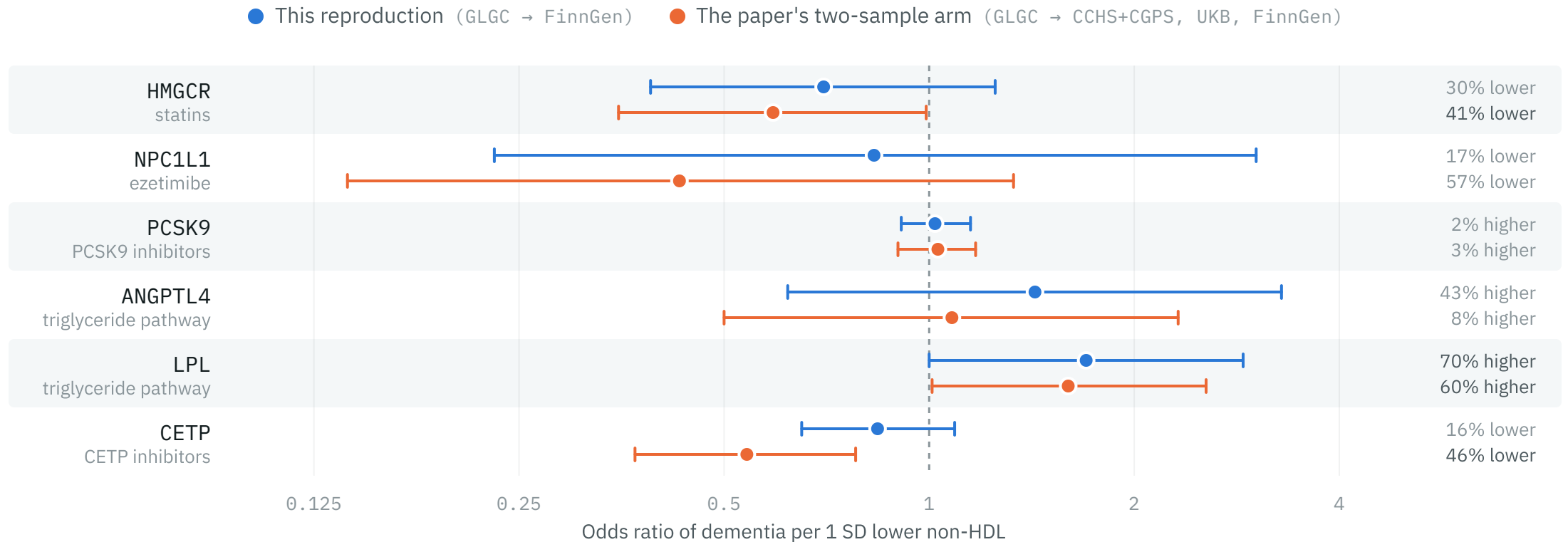
CLAUDE

- **Phase 0** — fetched the six gene regions only: ~10 MB, not 5 GB of whole-genome files
- **Phase 1** — six genes × two diseases, heart-disease check passing
- **Phase 2** — picked its own variants, five diseases, alternative estimators, assumption checks

Heart disease as a positive control



Our reproduction, against the paper's two-sample arm



Should we include UK Biobank participants in GLGC?

ALEX

Since we are not using uk biobank, does it make sense to use GLGC release that includes uk biobank participants?

CLAUDE

- **Argued to use release that excludes UK Biobank**
- Interval widths are driven by the low dementia numbers
- Introduces bias if we add a dementia dataset that includes UK Biobank participants

Asking what would actually move the result

ALEX

what is the easiest database to get access to to increase outcome numbers?

CLAUDE

- **OpenGWAS** — already readable by the tools we were using
- ~39,000 Alzheimer's cases against our 6,000

More cases, and what they moved

ALEX

lets proceed with opengwas

CLAUDE

ANGPTL4 → ALZHEIMER'S

6K CASES

~45K CASES

Odds ratio (95% CI)

2.56 (0.88–7.44)

2.62 (1.38–4.95)

Odds of Alzheimer's per 1 SD lower cholesterol through this target.

"Not significant" is two different results

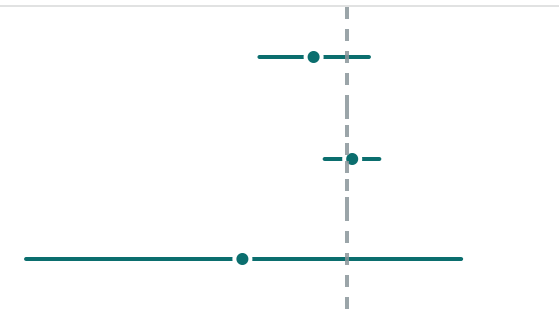
CLAUDE "... the remaining nulls are a power story ..."

ALEX

but didn't we add a lot of cases from opengwas so the power should be good no?

CLAUDE Conceded — for Alzheimer's that is no longer true:

TARGET → OUTCOME	CASES	ODDS RATIO (95% CI)
CETP → Alzheimer's	6,100	0.82 (0.58–1.17)
CETP → Alzheimer's	45,000	1.05 (0.88–1.25)
HMGCR → vascular dementia	2,700	0.52 (0.13–2.11)



The tool had instructions I had not found

ALEX

put together a report.Rmd with background and details about our reproduction...

CLAUDE

1 The paper's headline result
(not ours)

2 What is Mendelian
randomization?

Lipid drug-target genes & dementia risk — a two-sample MR reproduction

Code ▾

GLGC (exposure) × FinnGen (outcome) with an OpenGWAS extension

⋮

ALEX

recreate the report using the **artifact-design** and **dataviz** skills

CLAUDE

METHOD

Two-sample MR

GLGC × FinnGen, summary data only

POSITIVE CONTROL

✓ **PASS**

IHD protective for all 4 anti-
atherogenic targets

CONCORDANCE

6 / 6 targets

match the paper's own 2SMR direction

Claude wrote down a limit that did not exist

IN THE NOTES "unspecified dementia (ICD-10 F03) has no clean standalone R12 public endpoint"

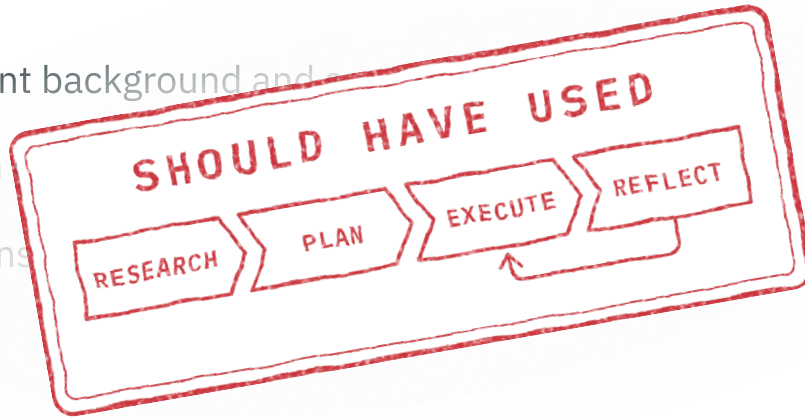
ALEX

wasn't FinnGen R12 used by the paper as well so shouldn't it be there?

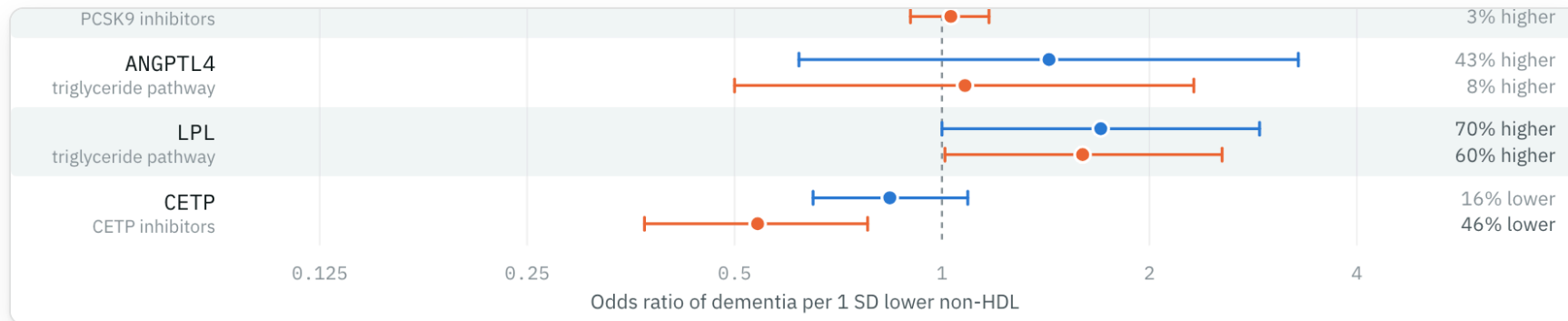
- CLAUDE
- Listed all endpoints — **F5_DEMNAS** is unspecified dementia, ICD-10 F03
 - Original search tried "UNSPEC" and "F03"
 - Added it and re-ran

Check all AI-generated text

- Insufficient background and
- Too much
- Overly sens



IN THE REPORT “... the lone real divergence is CETP ...”



● This reproduction ● The paper's two-sample arm

Review content before publishing or sharing

AI-generated content can be inaccurate, misleading, or entirely fabricated (sometimes called “hallucinations”) or may contain copyrighted material. **You are responsible for any content that you publish or share that includes AI-generated material.**

– HUIT guidelines on generative AI

HUIT huit.harvard.edu/ai/guidelines

HMS IT it.hms.harvard.edu/about/policies-and-guidelines/generative-ai-guidelines

Methodology > prompts

- **Gather relevant facts into research.md** — the source of truth for the plan
- **Align on a phased plan.md** — review this manually
- **Implement the plan**
- **Reflect means critically reading the output**